

Risk-Based Borrowing Limits in Credit Card Markets*

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Abstract

I use statement-level credit card data to show that UK lenders primarily tailor contracts through risk-based credit limits. This contrasts with US lenders, which personalize interest rates and credit limits. I estimate a structural model of the UK credit card market in which lenders set risk-based credit limits and interest rates under regulations that constrain rate personalization. Counterfactuals show that lenders earn 4% higher profits by personalizing interest rates rather than limits, and that tailoring interest rates in addition to credit limits raises profits by 12%. When interest rate personalization is constrained, credit limit flexibility offsets 46% of profit losses.

Keywords: Risk-based credit limits, risk-based pricing, adverse selection, credit cards

JEL Classification: D22, D82, G21, G28, G51, L13

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1 Introduction

Credit cards are a leading, heavily regulated source of unsecured consumer credit. Two key variables govern how credit card markets function: interest rates and credit limits. Policy-making in these markets focuses primarily on the former, with US and EU/UK governments introducing wide-ranging policies that restrict how lenders personalize interest rates, and recent academic work evaluating their impacts (Nelson, 2025). When lenders are restricted from personalizing interest rates, they may substitute toward other contractual terms, most naturally the credit limit. Yet no model of the credit card market endogenizes the credit limit alongside the interest rate, leaving open the extent to which one instrument can substitute for the other, the value of each instrument to lenders, and the overall impact of regulations limiting price personalization.

I develop and estimate a structural model of the UK credit card market in which lenders set both risk-based credit limits and interest rates in an oligopoly, under UK regulation that constrains interest rate personalization. With the estimated model, I can switch each instrument’s personalization on and off while lenders re-optimize the other, isolating each instrument’s contribution in equilibrium.

I use the estimated model to answer three questions about the effects of allowing interest rate and credit limit personalization. First, do lenders gain more from being able to personalize customers’ interest rates or credit limits? I find that lenders only marginally prefer to personalize interest rates rather than credit limits, with profits approximately 4% higher when only interest rates are personalized than when only credit limits are personalized. Second, does allowing personalized pricing alongside tailored credit limits increase or decrease lenders’ profits on average? Unlike a monopolist, an oligopolist’s profits may fall under price personalization, as competition for individual customers could erode the gains from price discrimination. Nevertheless, I find that on aggregate, lenders benefit from individualizing interest rates on top of credit limits, with average profits 12% higher when interest rates

can also be personalized costlessly. Third, how much of the effects of limiting interest rate tailoring can lenders offset by personalizing the credit limit margin? I find that lenders can recoup 46% of the profit losses from regulatory constraints on personalizing interest rates by tailoring credit limits. Hence, while policies restricting interest rate personalization prohibit the margin lenders value most, they recover much of the lost profit by adjusting on the credit limit margin.

To obtain these results, I draw on a new source of statement-level data covering approximately 80% of UK credit cards active in 2010–2015. The data include individual cardholder demographics and card characteristics, along with monthly spending, repayment, default decisions, and lenders’ funding costs. Among other advantages, the data include lenders’ proprietary credit scores for every card originated. Hence, I can assess whether lenders individualize interest rates and credit limits according to their predictions of customers’ risk. The data reveal widespread intensive-margin credit rationing: 40% of individuals use over 90% of their credit limit at least once in the first two years of owning the card.

The data also reveal extensive variation in credit limits, both within and across lenders. This variation persists within credit card products, with higher credit scores mapping to larger credit limits. In contrast, interest rates vary minimally across cards, remain nearly constant across customers at the card-month level, and are not risk-based. This pricing choice by lenders exceeds EU/UK regulatory requirements, which mandate that all credit products feature a representative APR that applies to at least 51% of customers each month.

The demand side of the structural equilibrium model I develop explains borrowers’ credit card choices, unpaid balances (borrowing), and default decisions. For customers’ borrowing choices, I allow preferences over interest rates to vary with income, introducing a potential motive for price personalization. I exploit a novel source of quasi-experimental price variation to identify demand elasticities: the cost shock resulting from the 2011 High Court ruling on the mis-selling of payment protection insurance (PPI). Banks were forced to compensate

thousands of consumers after the court deemed they had mis-sold PPI, and interest rates on some credit cards subsequently increased.

On the supply side, lenders choose each customer’s personalized credit limit and by how much to deviate from the advertised interest rate. Setting the credit limit trades off two forces: a larger limit raises interest revenue when the customer borrows and repays, but it also exposes the lender to adverse selection, because the unobservables driving a customer’s desired balance are positively correlated with their default probability, so that those with higher levels of borrowing are more risky on average. Lenders make their choices based on a noisy signal of each customer’s risk. The model also allows for a cost of individualizing a customer’s interest rate above the advertised level, reflecting frictions that could include infrastructure costs and the reputational risk of customer backlash.

I simulate outcomes in the estimated model under four scenarios: interest rates and credit limits both fixed at the card level; both interest rates and credit limits individualized without cost or constraint; and two intermediate cases in which exactly one margin can be personalized. Alongside the three headline results mentioned earlier, I find that the profit gain from adding interest rate personalization is smaller when credit limits are already personalized than when credit limits are constant. The two instruments are therefore substitutes rather than complements in profit, in that each instrument is worth less at the margin once the other instrument is already personalized.

My results relate to ongoing debates about the effectiveness of policy restrictions on personalized pricing. Prompted by the spread of algorithmic pricing, a growing number of US states have moved to restrict sellers’ use of consumers’ personal data to set individualized prices ([State of Maryland, 2026](#); [State of Connecticut, 2026](#); [State of New Jersey, 2026](#)). I show that in the credit card market, firms can substantially offset restrictions on personalization on one margin by re-optimizing another margin. These results imply that assessing the effects of price-personalization regulation on lenders requires accounting for the other

margins along which they might individualize contracts.

1.1 Related Literature and Contribution

Credit Limits and Credit Rationing. Credit limits are a device for rationing credit, and a long theoretical literature dating back beyond the seminal work of [Stiglitz and Weiss \(1981\)](#) highlights the role for controlling credit quantities (see [Calomiris et al. \(2017\)](#) for a review). The empirical literature on credit limits focuses on how credit limits affect borrower outcomes ([Gross and Souleles, 2002a](#); [Agarwal et al., 2017](#); [Aydin, 2022](#); [Yin, 2026](#)). Together, the evidence shows that higher credit limits can substantially increase borrowing. What I add to this literature is descriptive evidence and a structural model of how *lenders* set credit limits in the first place.

The state-of-the-art modeling of credit card markets is [Nelson \(2025\)](#), in which borrowers make dynamic discrete choices about their credit card product and *whether* to borrow, based on lenders' interest rate choices. In that model, there is no intensive-margin decision about how much to borrow, and the focus differs from how credit limits shape borrower behavior. A recent model of a different credit market featuring competition and screening over *multiple* contract terms is [Polo et al. \(2025\)](#), which focuses on menus of mortgage interest rates and loan-to-value ratios rather than lenders that individually tailor prices and/or credit quantities according to predictions of customers' risk. My contribution is to estimate a structural model in which credit card lenders endogenously set risk-based credit limits, a defining feature of credit card markets. My model is rich enough to enable counterfactual analysis in which lenders choose credit limits and interest rates that are either personalized for each customer or held constant at the card level, enabling me to explore how personalized prices and credit limits affect lenders in equilibrium.

Personalized and Risk-Based Pricing. The empirical literature documents substantial dispersion in interest rates across credit card borrowers in the US ([Galenianos and Gavazza, 2022](#); [Nelson, 2025](#)). Separately, [Livshits et al. \(2016\)](#) argues that credit scoring and risk-

based pricing “democratized” US credit, with increased access to credit coinciding with rising rate dispersion. This variation contrasts with the UK, as I show in this paper, where credit is widely available across the risk distribution but rates vary little, consistent with risk-based credit limits substituting for risk-based rates. There is limited other work documenting variation in credit limits and interest rates in European consumer credit markets.

More broadly, risk-based pricing is one instance of *personalized pricing*, for which there is a large theoretical literature (Thisse and Vives, 1988; Corts, 1998; Holmes, 1989) and growing empirical evidence (Dube and Misra, 2023; Durrmeyer et al., 2026). A central message from these papers is that the profit effect of personalization of prices is not obvious *a priori*: Rhodes and Zhou (2024) shows that the sign depends on how much of the market is already served; where most consumers already buy at a common price (high coverage), personalization intensifies the competition for them and can lower average profits. Bergemann et al. (2015) characterizes how widely price discrimination can redistribute surplus depending on the information available to firms. My model allows me to quantify whether personalized pricing and credit limits increase profits, and which margin of flexibility lenders value more. Other literature on the effects of personalized credit limits on lenders is limited.

Regulation of Credit Card Markets. An empirical literature studies the regulation of credit card markets, with much of it centered on the 2009 US CARD Act: Agarwal et al. (2014) finds large consumer savings from the Act, while Nelson (2025) studies how it restricted lenders’ ability to reprice customers after origination. Because such regulation mainly constrains interest rates, the credit limit becomes a potentially valuable margin over which lenders retain discretion, motivating a model in which they are set endogenously. My contribution to this literature is to provide such a model and quantify how regulations on personalized interest rates impact consumers and lenders, in the context of personalized credit limits.

2 Data, Summary Statistics, and Institutional Context

My primary data source is the Financial Conduct Authority (FCA) Credit Card Market Study (CCMS) Database (FCA, 2015a). The FCA used its regulatory authority to collect data on all credit cards active between 2010 and 2015 at 14 lending entities, representative of the UK credit card market. The dataset covers approximately 74 million cards, representing about 80% of all UK cards active during this period. The FCA’s data collection was organized into three main datasets, none of which have been used for academic research.

The first dataset includes cardholder and card information at origination, such as demographics (e.g., age, income, employment status), acquisition channel (e.g., in-branch, online, by post), and initial interest rate, credit limit, and length of any 0% promotional deal. This dataset also includes lender-specific credit scores, which, in the absence of a UK FICO equivalent, are needed to assess whether interest rates and credit limits are risk-based. Table A.1 provides summary statistics for the datasets. The statistics reveal that the coefficient of variation for credit limits is 93%, compared with 36% for APRs, implying much less variation in interest rates than in credit limits. Sections 3.1 and 3.2 further decompose these variations in interest rates and credit limits within lenders across card products and within card products across individuals.

The second dataset is monthly panel data of credit card statement information from January 2010 to January 2015. It details opening and closing balances, repayments, the number and value of transactions, fees, interest, and failures to repay. These data allow me to measure *utilization* (balance used up to the closing date) and *revolving* (balance left on the account after repayment). The dataset also includes the evolution of credit scores, interest rates, and credit limits. Following industry practice, I use observations on failures to repay to classify customer default as 90+ days without a repayment. These serious delinquencies occur for just under 4% of customers in the first 18 months of card ownership.

The statement data show that credit rationing, in the form of near- or full-credit limit

utilization, is a prominent feature of the UK credit card market.¹ Among statements with a positive closing balance, 23% close with a balance over 90% of the credit limit, and 27% close with a balance exceeding 85% of the limit. At the individual level, in the first two years after origination, approximately 40% use more than 90% of their credit limit at least once.

Repaying the entire balance—also known as transacting—is another prevalent feature of the statement data. Repayment covers the full balance on about 50% of statements, and about 25% of cardholders repay in full every month in the first 12 months post-origination. The proportion of transacting statements varies substantially across lenders, ranging from 22% to 85%. My model includes a consumer choice about whether to transact and lender-specific variation that generates sorting along this choice.

The third dataset is a monthly panel of card characteristics covering January 2010 through January 2015, capturing annual fees, rewards, income thresholds, and advertised APRs. The data also include lender funding costs. The summary statistics on funding rates relative to advertised APRs in Table A.1 indicate that credit card lenders earn substantial markups, as also shown for US consumer (Herkenhoff and Morelli, 2026) and business (Benetton and Buchak, 2024) credit cards.² That the marginal lender sets interest rates above the marginal cost of lending motivates modeling the credit card supply side as imperfectly competitive.

The CCMS data package also includes a credit reference agency dataset that matches cards to individuals. This is an advantage of the data over the Federal Reserve’s Y-14M US credit card data, which precludes matching cards to individuals across lenders (examples of papers using the Y-14M include Bord et al. (2025), Drechsler et al. (2025), and Berger et al., 2024).

¹Individuals using, say, 85% of their credit limit are considered credit constrained because they might not make a purchase on credit because it would push them over their credit limit, which incurs a fee and will lead to a decrease in credit score.

²The mean yearly funding rate is 2.28%, and fewer than 4% of borrowers default each year. Even under the conservative assumption that lenders cannot recover any portion of a defaulted balance and ignoring interchange revenue, this implies a markup (price-to-marginal-cost ratio) of over 3.

UK Credit Card Market and EU Regulation. The UK credit market differs from the US market in several respects. First, rewards, purchase promotional deals, and fees are far sparser in the UK, with 88% of card-month observations having no annual fee and only 6% of cards featuring air miles rewards. Second, US individuals own more credit cards (2.24 on average) than UK individuals, who typically have one.

Most important for what follows is how interest rates must be advertised and disclosed. All promotional material and documentation for a credit card product in an EU credit market must include a “representative” (“advertised”) APR, and a specified share of customers each month must receive that advertised rate or lower—at least 66% before February 2011, and at least 51% since, when the UK harmonized with the EU Consumer Credit Directive. The US has no such regulation on interest rates at origination, though the CARD Act limits dynamic repricing of customers, something that is not regulated in the UK. Two other features of the UK market worth noting are that (i) UK customers are not able to see their personal interest rate or credit limit until after they are approved, and (ii) the UK has no binding laws capping the level of credit card interest rates. While some US states have usury laws, these caps tend not to apply because national banks are regulated at the federal level.

3 Uniform Interest Rates, Risk-Based Credit Limits

The aim of this section is to present robust evidence that leading UK credit card lenders individualize credit card contracts through risk-based credit limits rather than interest rates. Although not described in detail below, it is worth noting that a card’s rewards, fees, and promotional deals are also constant across customers, implying that the *only* variable personalized to a meaningful extent is the credit limit.

3.1 Interest Rates

I start by documenting limited variation in interest rates across customers at each lender. Table A.2 reports the average (over months) of lenders' interest rate coefficient of variation. The values are below 23%, and the average for prime and superprime lenders (weighted by originations) is 14%. Further, the across-lender weighted average of the ratio of the 75th to 25th percentiles (respectively, the 90th to 10th) for interest rates is 1.19 (respectively 1.38), further illustrating limited variation in interest rates within UK lenders. In contrast, Galenianos and Gavazza (2022) show that for US interest rates, the ratio of the 90th to the 10th percentile is as large as three, even after controlling for borrower and card characteristics.

Among the leading UK credit card lenders, a modest share of the already small total variation in interest rates at a lender is across individuals on a given credit card product. I illustrate this feature by performing a one-way Analysis of Variance (ANOVA) that decomposes the variation in interest rates at a lender into a within-card and between-card component (see Appendix B.1 for the mathematical formulation). Table A.2 reports the percentage of within-card variation for each of the lenders. On average, within-card variation for prime and superprime lenders accounts for only 24% of the total variation. There is also one lender that gives all customers on a given credit card the same interest rate in all origination months in the data.

Related to the lack of within-card variation in interest rates, I calculate the percentage of customers in each month who obtain their card's advertised APR (recall that regulations require lenders to advertise an APR for each card product). I plot the time series in Figure A.1. The proportion of customers receiving the advertised APR across all credit cards in the sample remains consistently high at 80 to 90%, and this stability persists even after the regulatory change in February 2011 that relaxed the requirements for advertised APRs. Further, in 81% of card-months, over 90% of originations obtain the advertised APR. This statistic confirms that most *cards*, not just *lenders*, give most of their holders the advertised

APR. The lack of variation from the advertised APR implies there is no meaningful risk-based pricing for the vast majority of customers, which contrasts with the US evidence in Nelson (2025), where interest rates vary smoothly from 20% down to 12% across the FICO score distribution. The UK results suggest that either the benefits of individualizing interest rates are small or that lenders face significant shadow costs of individualizing rates. The model I estimate and the counterfactual analyses I conduct explore these benefits and costs.

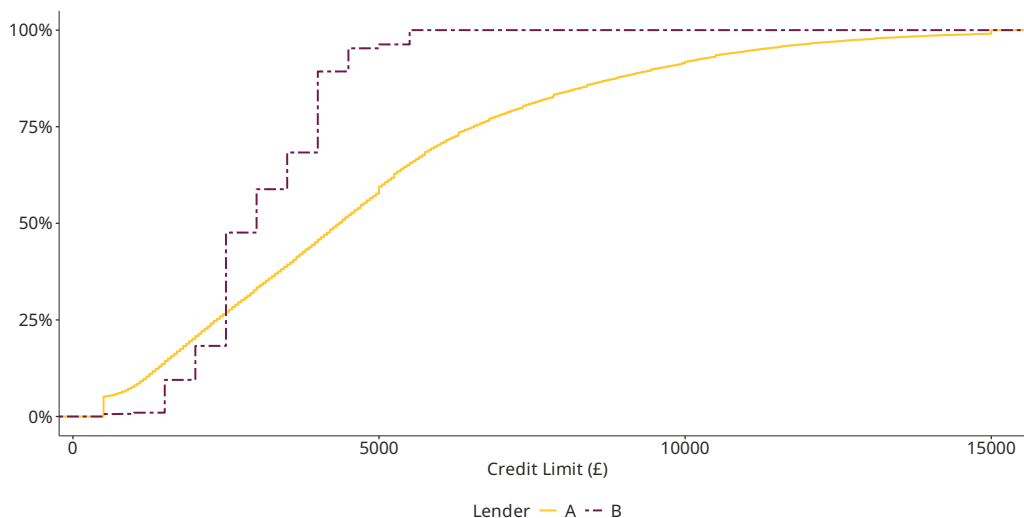
Lenders could implement risk-based pricing through two alternative channels: (i) dynamically repricing customers after origination, or (ii) sorting customers onto cards with different interest rates. Evidence in Appendix B.3 rules out both: lenders offer a small portfolio of cards with similar interest rates and keep 95% of customers' interest rates constant for the first year post-origination. The Appendix also presents brief empirical evidence inconsistent with collusion, another possible explanation for the absence of individualized interest rates.

3.2 Credit Limits

Unlike interest rates, credit limits vary substantially across customers. The origination-weighted coefficient of variation across lenders is approximately 77% on average, more than five times the equivalent value for interest rates. Table A.2 reports across-lender weighted 75/25 and 90/10 percentile credit limit ratios of 3.34 and 9.18, indicating significant within-lender variation. As before, I decompose each lender's credit limit variation into within-card and between-card components. Around 80% percent of total variation is found within product. The dominance of within-variation suggests that lenders do not sort individuals onto cards with varying average credit limits. Instead, credit limits vary considerably across customers, even within a credit card product and month.

The distribution of credit limits varies substantially across lenders. I illustrate this in Figure 1 by plotting the empirical cumulative distribution function (CDF) of credit limits for two contrasting lenders, A and B. Two substantial differences are evident. The first concerns the

FIGURE 1. EMPIRICAL CDFs OF TWO PARTICULAR LENDERS' CREDIT LIMITS

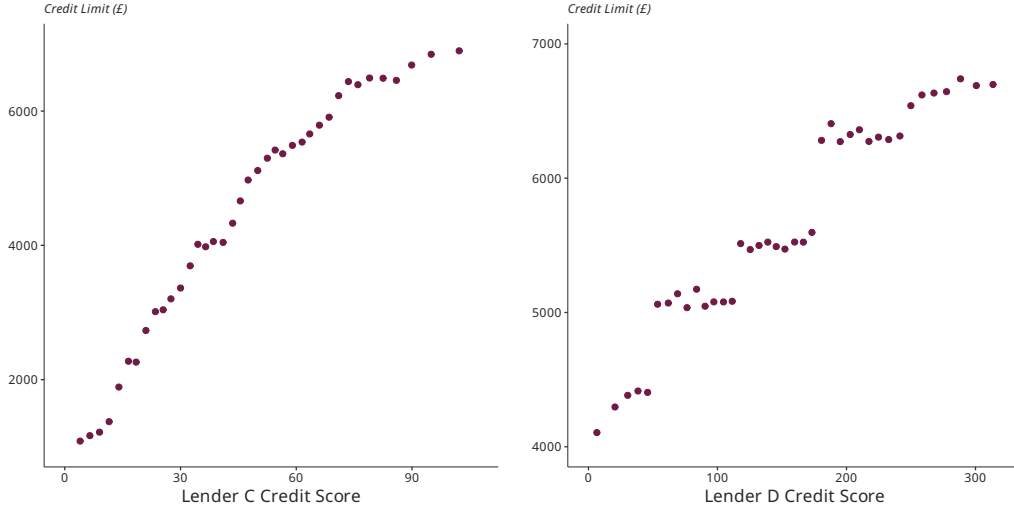


Notes: Monetary values here and everywhere that follows are quoted in 2015 Great British Pounds (GBP). Other lenders' CDFs are illustrated in Figure A.2.

shape of the distribution. Lender B's curve is step-like, consistent with a coarse process of assigning credit limits to individuals, in which groups of consumers receive identical credit limits. Lender A's smooth curve is consistent with a more finely tuned allocation mechanism for origination credit limits. Other lenders' CDFs (shown in Figure A.2) interpolate between these two extremes, varying in their coarseness. The second difference concerns the *scale* of credit limits. Lender A has lower values of credit limits than Lender B for the first 25 percentiles; however, all percentiles after the 25th are larger. The mean and range of lender A's credit limit distribution are indeed much larger. More formally, Anderson-Darling and rank-score k-sample tests reject equality of empirical CDFs across lenders at 0.5% significance.

As expected, lenders link each individual's credit limit to an assessment of the customer's default risk. In Figure 2, I plot the mean origination credit limit by application credit score for two contrasting lenders, C and D. The plot is produced for the year 2013, in which credit scores at the two lenders are stable. Both curves are upward-sloping, consistent with risk-based credit limits. Further, the right-hand lender exhibits discontinuities in credit scores

FIGURE 2. MEAN CREDIT LIMITS ACROSS CREDIT SCORES FOR TWO LENDERS



Notes: The plot is for the year 2013, in which the credit score measures at the lenders in the sample do not change. Credit score scales differ across lenders, so they cannot be compared. Figure A.3 contains the same plots for all lenders.

at credit score thresholds similar to those exploited in Agarwal et al. (2017). In Appendix Section B.5, I exploit such a discontinuity in credit limits by credit scores in a regression discontinuity design (RDD) to show that individuals do not adjust their card choice in response to their credit limit. This fact will be an important feature of the model I estimate.

3.3 Implications of Descriptive Findings and Additional Analysis

This section has shown that UK credit card lenders personalize credit limits based on assessments of customer risk but do not individualize interest rates. The next step is to understand how lender heterogeneity and the regulatory environment governing interest rate personalization affect equilibrium outcomes. The empirical setting alone is not informative about the impacts of the restrictions on tailoring interest rates, nor can it explain how lenders would set interest rates and credit limits if they were not required to set and advertise a card-level APR, or if they had to set a card-level credit limit instead of, or in addition to, advertised APRs. In the absence of meaningful exogenous variation in the regulatory environment or

the makeup of lenders, I build and estimate a model of the credit card market next.

Appendix B presents additional descriptive and quasi-experimental analyses. These analyses show that (i) lenders form their own risk scores and do not sort customers onto cards based on these scores (Section B.2); (ii) positively correlated unobservables drive borrowing and default decisions (Section B.4); and (iii) lower credit limits do not cause individuals to cancel their card, switch cards, or add another credit card to their portfolio (Section B.5). All these facts motivate the choices I make in the model.

4 Model of the Credit Card Market

In this section, I develop a model of lenders’ interest rate and credit limit choices and consumers’ credit card, revolving, and default choices. The demand model is relatively standard and builds on Crawford et al. (2018). The modeling contribution lies on the supply side, where lenders’ credit limit choices are endogenized as a function of lender-specific credit scores, and the costs and constraints in individualizing interest rates are parameterized. The setup lets me evaluate how restrictions on personalizing interest rates and credit limits affect lenders and borrowers, and compare the instruments from the lenders’ perspective.

4.1 Preliminaries and the Credit Card Product

The market is a pair (m, t) , where t denotes an origination month and m denotes one of three distribution channels: branch, online, or other channels (e.g., telephone or post). Origination months run from January 2010 to June 2013. I stop the sample at June 2013 so that I observe 18 months of borrowing and default behavior for each individual. With three distribution channels and 42 origination months, I have 126 markets in total.

The model has three components: credit cards $j \in J_{mt}$, consumers $i \in I_{mt}$ (demand), and lenders $\ell \in L_{mt}$ (supply). I model a credit card product as a bundle of features à la Lancaster (1966), a decision supported by Evans and Schmalensee (2005). The product-specific features

include the advertised interest rate r_{jmt} , the minimum income threshold $\underline{Y}_{jmt}$, observed characteristics X_{jmt} , and unobserved characteristics ξ_{jmt} (all described below). The credit limit is not among these features: unlike interest rates, there is no advertised credit limit.

4.2 Demand Model

I model consumers who do not hold a credit card at the point of shopping. Because most UK adults hold only one card (see Section 2), this is the empirically relevant population. Restricting to this population also circumvents complications from (i) balance transfers and (ii) balance-matching heuristics in repayment across multiple cards (Gathergood et al., 2019).

Card Choice (Shopping). Consumers choose a card and decide at origination whether to pay the balance in full (transacting nest) or leave some balance unpaid, accruing interest (revolving nest).³ That consumers fix this intent at origination is one of the few substantive restrictions I place on consumer behavior, as it greatly simplifies the lender’s problem.⁴

The consumer’s utility from revolving is $V_{ijmt}^E = \bar{V}^E(X_{jmt}^E, \xi_{jmt}^E, r_{jmt}, \eta_{mt}^E, y_i; \theta_{mt}^E) + \nu_{ijmt}$ where η_{mt}^E is a market fixed effect, y_i is the log of income Y_i , and θ_{mt}^E collects market-specific utility parameters. Superscript E denotes the Extensive margin, and X_{jmt}^E (e.g., rewards), ξ_{jmt}^E (e.g., brand loyalty) are the elements of X_{jmt} and ξ_{jmt} entering card choice. I take the taste shocks ν_{ijmt} to be generalized type-1 extreme value, independent across customers and correlated across choices within a customer, with nesting parameter ϱ_{mt} .

A key modeling choice is to exclude the individualized credit limit $\bar{b}_{ijmt}$ from $\bar{V}^E$. Three facts

³Treating transacting as the outside nest is a modeling device: it stands in for holding a card without borrowing on it, and accommodates the large share of transactors without separately modeling the decision to hold a card at all (on which I have no data). The maintained assumption is that a revolver facing less attractive card terms substitutes into transacting or another card, rather than ceasing to hold a card or moving the balance to a non-card product (e.g., personal loan or Buy Now Pay Later).

⁴Direct-debit take-up supports the assumption. Within six months of origination, 35% of consumers set up a direct debit. Of those doing so at origination, roughly 40% pay the full balance automatically (consistent with transacting), while most of the remainder pay only the minimum (consistent with revolving).

support this decision. First, credit limits are essentially never cited in answers to a question in the CCMS cardholder survey on which factors affected card choice (FCA, 2015b). Second, as an institutional fact, $\bar{b}_{ijmt}$ is set only after approval, so no individualized limit exists when consumers choose their card. What a consumer could plausibly act on at the shopping stage is a card's *expected* limit (e.g., if consumers favor cards they believe offer higher limits on average), and this channel is absorbed by ξ_{jmt}^E , which can include the card-average limit. The remaining question is whether expected variation in the limit, around the card average, shapes behavior.

My third fact addresses this question using an RDD, which I summarize here (see Appendix B.5 for a full discussion of the design, estimates, and robustness checks). I take a lender whose credit limits jump discontinuously at a risk-score threshold. The effects of a £750 (\$1200 in 2015 USD) discontinuous change in the credit limit on canceling, switching, and originating extra cards are small and statistically indistinguishable from zero. The estimates suggest that, on average, consumers do not meaningfully adjust their portfolios in response to unexpectedly low realized limits. While these findings do not perfectly identify initial shopping behavior as a function of credit limits, they provide strong supporting evidence for excluding individualized credit limits from the initial card-choice model.

I linearize card-choice utility as

$$V_{ijmt}^E = \beta^{E'} X_{jmt}^E + \xi_{jmt}^E + \alpha_{imt}^E r_{jmt} + \eta_{mt}^E + \nu_{ijmt}. \quad (1)$$

The coefficient α_{imt}^E is potentially individual-specific, capturing heterogeneous shopping-stage preferences over interest rates, with $\alpha_{imt}^E = \alpha^E + \Omega_{mt}^{E,r} \tilde{y}_{imt}$, where $\tilde{y}_{imt} = y_i - \bar{y}_{mt}$ is log income centered on the market average and $\Omega_{mt}^{E,r}$ governs how shopping-stage rate sensitivity varies with income.

In the baseline model, I set $\Omega_{mt}^{E,r} = 0$, so that shopping-stage rate sensitivity is homogeneous ($\alpha_{imt}^E = \alpha^E$ for all i). This choice ensures that the model does not artificially bake in interest

rates as a preferable tool to credit limits for lenders when tailoring contracts (since a primary question of the counterfactual analysis is which tool lenders value more). With $\Omega_{mt}^{E,r} = 0$, neither interest rates nor credit limits can sort consumers by income at the shopping stage, and any incentive to *tailor* either instrument arises *after* approval, at the intensive margin. This does not mean advertised rates and average credit limits leave market shares unaffected; it means deviations from those averages have no effect at the shopping stage. In Section 6, I discuss the results from a more general specification with $\Omega_{mt}^{E,r} \neq 0$ and more restrictive random-matching models in which interest rates play no role at the shopping stage ($\alpha^E = 0$ and separately $\bar{V}^E = 0$).

I generate choice sets for individuals by comparing their income at origination to the card’s income threshold, $\underline{Y}_{jmt}$. The set of cards available to customer i is $J_{imt} = \{j \in J_{mt} \mid Y_i > \underline{Y}_{jmt}\}$. Income thresholds allow lenders to meet affordability and creditworthiness regulatory requirements (FCA, 2024).

The utility from transacting is given by $V_{i0mt}^E = \delta_{0mt} + \nu_{i0mt} + \Omega_{mt}^{E,\text{cons}} \tilde{y}_{imt}$, where δ_{0mt} is a market-level transacting constant. A consumer chooses the card j^* that maximizes V_{ijmt}^E in their choice set, and transacts if V_{i0mt}^E exceeds $V_{ij^*mt}^E$.

Revolving (Borrowing). Revolvers choose how much of their initial balance to leave unpaid, a decision interchangeably referred to as revolving and borrowing.⁵ Let b_{ijmt}^* denote the *desired* level of revolving, the amount an individual would leave unpaid on their card, absent any credit limit. The word “desired” reflects that they may wish to borrow more than

⁵Revolvers are likely to make monthly borrowing choices, such as those implied by the solution to an intertemporal consumption-savings problem. However, this paper concerns lenders’ choices of *origination* credit limits and interest rates. When choosing origination credit limits, lenders care more about consumers’ overall borrowing over the immediate period they use the card than about the dynamics of borrowing within that period. After all, lenders can reprice at any point after origination if they desire. Hence, my setup does not necessitate a model of multiple borrowing values across periods, as a consumption-savings problem implies. More generally, my demand model of card origination, revolving, and default can be microfounded in a typical consumption-savings lifecycle setup. However, since my focus is lenders’ credit limit and interest rate choices, I prefer to specify demand-side estimating equations as a set of linearized equations agnostic to the behavior that generates them.

their limit $\bar{b}_{ijmt}$ allows. Desired revolving is $b_{ijmt}^* = b(X_{jmt}^B, \xi_{jmt}^B, r_{jmt}, \eta_{mt}^B, y_i, \varepsilon_{imt}^B; \theta_{mt}^B)$, and observed, constrained revolving $b_{ijmt} = \min\{b_{ijmt}^*, \bar{b}_{ijmt}\}$. The terms X_{jmt}^B , ξ_{jmt}^B , and η_{mt}^B in (2) have analogous definitions to those in (1), with E replaced by **B**orrowing. Borrowing is written as a function of the advertised rate rather than the individualized rate (see Section 5.1 for justification of this choice). The term ε_{imt}^B reflects an unobserved demand for borrowing. For example, ε_{imt}^B would be high if an individual has an unreported health issue that requires them to separate from their job. Neither the lender nor I observe ε_{imt}^B perfectly. Finally, I take the log of initial borrowing to be linear in parameters:

$$\log(b_{ijmt}^*) = \beta^{B'} X_{jmt}^B + \xi_{jmt}^B + \alpha_{imt}^B r_{jmt} + \eta_{mt}^B + \Omega_{mt}^{B, \text{cons}} \tilde{y}_{imt} + \varepsilon_{imt}^B, \quad (2)$$

where, similar to before, $\alpha_{imt}^B = \alpha^B + \Omega_{mt}^{B, r} \tilde{y}_{imt}$.

Default. Revolvers choose whether to default. The net utility from defaulting is

$$V_{imt}^D = V^D(\eta_{mt}^D, y_i, \varepsilon_{imt}^D; \theta_{mt}^D) = \eta_{mt}^D + \Omega_{mt}^D \tilde{y}_{imt} + \varepsilon_{imt}^D, \quad (3)$$

where the terms carry their earlier meaning, with E replaced by **D**efault. The individual defaults if $V_{imt}^D > 0$. I follow the empirical literature on credit markets and omit the interest rate from default utility (see, e.g., Nelson, 2025). I also assume default is not a direct function of the credit limit (see Gross and Souleles (2002b) and Aydin, 2022). This makes the lender credit limit problem one of selection, since absent a direct channel, a larger limit raises default only by attracting riskier borrowers. Appendix C.2 discusses both assumptions and the supporting evidence, and shows that the association between credit limit and default in my data is consistent with a muted positive causal effect of the limit on default.

Private Information. The revolving and default unobservables share a common component ε_i :

$$\varepsilon_{imt}^B = \sigma_{mt}^B \varepsilon_i, \quad \varepsilon_{imt}^D = \sigma_{mt}^D \varepsilon_i + \tilde{\varepsilon}_i^D,$$

with $(\varepsilon_i, \tilde{\varepsilon}_i^D)$ independent standard normal. The common component links the two unobservables, enabling intensive-margin selection, and simplifies the lender's signal structure that

follows. Including either $\tilde{\varepsilon}_i^B$ or a term ε_i^E in the transaction utility would be superfluous, and both are excluded. Regarding ε_{imt}^E , it would only affect the decision between transacting and revolving—it would drop out of decisions *between* card products—and therefore, its role is already subsumed by ε_{imt}^B .

4.3 Supply Model

Preliminaries. Lenders observe customers’ incomes Y_i and treat the product menu (i.e., characteristics X_{jmt} and ξ_{jmt} , and income thresholds $\underline{Y}_{jmt}$) as given in each market. Three factors support this latter choice. First, rewards are constant across individuals and sticky, rarely changing over the five-year period I observe. Second, lenders cannot adjust most unobserved characteristics, such as loyalty, within a given month. Third, taking the menu as given yields a tractable, estimable problem in credit limits and interest rates, avoiding issues with equilibrium existence and uniqueness (at the cost of the model not endogenizing how lenders choose their product set).

Each lender ℓ screens applicants using its own credit scoring technology, which produces a tailored distribution over a customer’s common risk component ε_i (whose population distribution is standard normal). The technology, treated as exogenous, is characterized by a finite set of signals $\{e_{\ell t 1}, \dots, e_{\ell t L_{\ell t}}\}$ and a dispersion term $\sigma_{\ell t}^2 \leq 1$. For an individual who generates signal $e_{i \ell t}$, the technology generates the predicted risk distribution $\mathcal{N}(e_{i \ell t}, \sigma_{\ell t}^2)$. The interested reader can find further details on the modeling of the screening technology, an illustrative example, and a comparison to the existing literature in Appendix C.3.

Timing. The timing follows the institutional environment. Lenders set advertised APRs for each card at the start of each month; individuals then choose a card; lenders obtain signals on their customers, and only *after* individuals sign their credit agreements do lenders set potentially individualized credit limits and interest rates, subject to the regulatory requirements. Advertised interest rates are chosen oligopolistically, but because lenders choose credit limits and individualized interest rates after consumers select their card, they are set

non-strategically to maximize profits, with no “winner’s curse” (Shaffer, 1998).

For further justification of this modeling choice, note that the 1974 Consumer Credit Act mandates a “cooling-off period” during which consumers can cancel their card, though my data show that this option is exercised in only 0.2% of originations, implying that borrowers do not return to shopping after interest rates and credit limits are revealed.

Advertised Interest Rates. Lenders first choose r_{jmt} to maximize expected profits so that interest rates form a Nash-Bertrand equilibrium. For lender ℓ , their vector of advertised rates $\mathbf{r}_{\ell mt}^*$ maximizes expected profits taking rivals’ equilibrium rates $\mathbf{r}_{-\ell mt}^*$ as given.

Individualized Interest Rates and Credit Limits. Lenders then set individual-specific credit limits $\bar{b}_{ijmt}$, and also individual-specific interest rates r_{ijmt} by choosing deviations from the advertised rate $z_{ijmt} = r_{ijmt} - r_{jmt}$. The origination data show that z_{ijmt} is zero for about 90% of originations, and when z_{ijmt} is nonzero, at prime and superprime lenders it is positive in 98% of cases. I therefore focus on the empirically relevant case $z_{ijmt} \geq 0$, noting that regulation requires z_{ijmt} to be non-positive for at least 51% of a card’s originations within a month (66% before February 2011).

Lenders’ Profits and Optimization Problem. Modeling lenders’ credit limit and tailored interest rate choices requires an expression for their profits from each customer. For costs, I focus on the cost of funds, c , charge-off (default) costs, and the shadow costs of individualizing interest rates, C . In the US, the first two account for over two-thirds of lenders’ total cost of issuing credit cards (Evans and Schmalensee, 2005). The remaining third comprises mainly fixed costs (overhead and operational), which are sunk when lenders set each customer’s rates and limits. Regarding revenue, I focus entirely on finance charges coming from interest. For US lenders in 2001, this accounted for 70% of their card revenue. The remaining 30% comes from interchange revenue, fees, and cash advances. Appendix C.4 describes the three factors in more detail and explains why they are less prominent in the UK credit card market relative to the US. Finally, a lender’s profit from a transacting customer, Π_{i0mt} , does not depend on the credit limit and interest rate; thus, credit limit and individ-

ualized interest rate decisions are unaffected by whether the individual originating the card is a transactor or a borrower.⁶

The total expected profit *per unit credit* revolved for borrower i on card j is given by

$$\pi_{ijmt} = (1 - \Delta_{imt})(r_{ijmt} - c_{jmt}) + \Delta_{imt}(-1 - c_{jmt}), \quad (4)$$

where Δ_{imt} denotes the probability that borrower i defaults and c_{jmt} the funding rate. This expression imposes that lenders recover none of a charged-off balance.⁷ The expected profit from borrower i choosing card j in market (m, t) , *gross* of any costs of individualizing interest rates, is

$$\Pi_{ijmt} = \mathbb{E} [\min\{b_{ijmt}^*, \bar{b}_{ijmt}\} \pi_{ijmt}], \quad (5)$$

where the expectation is over the distribution $\mathcal{N}(e_{i\ell t}, \sigma_{\ell t}^2)$ from the screening technology. For card j , across their originating customers $i \in I_{jmt}$, the lender solves

$$\max_{\{\bar{b}_{ijmt}, z_{ijmt}\}} \sum_{i \in I_{jmt}} \Pi_{ijmt}(\bar{b}_{ijmt}, z_{ijmt}) - C(z_{ijmt}; \kappa_{ijmt}),$$

subject to $z_{ijmt} \geq 0$ for all i and the regulatory constraint

$$\frac{1}{|I_{jmt}|} \sum_{i \in I_{jmt}} 1(z_{ijmt} > 0) \leq \chi_{\ell t}, \quad (6)$$

where I_{jmt} denotes the set of revolvers choosing card j in distribution channel m and month t , $1(A)$ denotes the indicator function, equal to 1 if A is true and 0 otherwise, and $\chi_{\ell t}$ is

⁶The revenue and costs from transactors do not relate to the interest rate, since they do not revolve a balance on which interest accrues. Lenders' variable cost from non-defaulting customers is per-unit credit; therefore, lenders' costs from transactors are unrelated to the credit limit. Limits may affect interchange revenue, but I abstract from interchange revenue for revolvers and transactors for the same reason.

⁷When cardholders default, payment card issuers start collection procedures. These cardholders will often have other debts, which are typically collected before credit card debt. Debt collection procedures are very costly relative to the size of the loan for credit card lenders. Further, in the US in 2002, 50% of all charge-offs resulted from bankruptcy, where debt collection is often futile (Evans and Schmalensee, 2005).

the regulatory requirement on the proportion of revolvers receiving a deviation from the advertised rate.

The term $C(z_{ijmt}; \kappa_{ijmt})$ is the cost to the lender of deviating by z_{ijmt} from the advertised rate, given the parameters κ_{ijmt} . It is increasing and weakly convex in z_{ijmt} and satisfies $C(0; \kappa_{ijmt}) = 0$. My context does not allow me to identify the exact sources of C , rather, the goal is to quantify its magnitude off lenders' optimal choices. Nevertheless, interviews with industry experts pointed to two main candidates: reputational costs and the administrative expenses (infrastructure, software, etc.) of implementing optimal individualized prices (see Appendix C.5 for details).

The Credit Limit Optimality Condition. As derived in Appendix C.6, the first order condition with respect to the credit limit is

$$\mathbb{E} [\pi_{ijmt} | b_{ijmt}^* \geq \bar{b}_{ijmt}] = 0. \quad (7)$$

The intuition behind the first order condition is that, at the optimal credit limit, the expected profit per unit credit, over those with unobservables that drive them to use their full credit line, is zero. If the expected profit per unit credit were instead positive, the lender should raise the credit limit because the expected benefit of safer types using the entire credit limit exceeds the expected costs of riskier types using the whole credit limit (the reverse argument holds if expected profit per unit credit were negative). The first order condition here is not a zero-profit condition: expected profit over those infra-marginal individuals with unobservables that drive them to use less than their full credit line is positive.

The economic force behind the optimality of this condition is a positive correlation between the unobservables driving borrowing and default, which I call intensive-margin adverse selection: borrowers who exhaust their credit line have the strongest unobserved demand to borrow and, given this correlation, are the most likely to default. Lenders therefore cannot expand a limit to safer borrowers without simultaneously extending credit to the riskier

types who exhaust it—this tradeoff pins down the optimal limit. Existing studies of credit markets estimate adverse selection along this margin (Crawford et al., 2018; Benetton and Buchak, 2024). My demand estimates, which I obtain entirely *independently* of the supply side, verify this assumption (see Sections 5 and 6 for details).

Because default risk does not depend directly on borrowing, this selection also ensures the lender’s problem is well-behaved. Intuitively, conditional on observables, only the highest unobserved-risk borrowers—those with negative expected per-unit profit—utilize sufficiently large limits, and the lender does not wish to extend them further credit; this is what bounds the optimal limit. Without the selection effect, the problem would have a bang-bang solution: lenders would provide zero credit to borrowers with negative expected per-unit profit and unbounded limits to those with positive per-unit profit. To summarize, in the context of adverse selection, the choice of credit limit must be made with consideration of the distribution of default risk *among those* utilizing that amount of credit. This insight reveals the effect of adverse selection on lenders’ optimal credit limit choices.

Individualized Interest Rate Optimality. Since the regulatory constraint in Equation (6) is slack for every lender, the optimality condition for z_{ijmt} is

$$\frac{\partial \Pi_{ijmt}}{\partial z_{ijmt}} = \frac{\partial C}{\partial z_{ijmt}} - \lambda_{ijmt}, \quad (8)$$

where λ_{ijmt} is the multiplier on $z_{ijmt} \geq 0$. For individuals with $z_{ijmt} > 0$ (and thus $\lambda_{ijmt} = 0$), the marginal increase in profit from changing z_{ijmt} , i.e., $\partial \Pi_{ijmt} / \partial z_{ijmt}$, must equal the marginal cost of individualizing interest rates $\partial C / \partial z_{ijmt}$.

5 Model Estimation

In this section, I summarize my method for estimating model parameters; Appendix D provides further details. I focus here on how I overcome the main challenges in estimation

and discuss the exogenous variation I exploit to identify the parameters. Figure A.4 illustrates the five steps of the estimation procedure, which I describe in turn.

5.1 Demand Estimation

Log-likelihood Conditional on Revolving. The first step estimates the first set of parameters of the demand model. My demand model consists of equations for revolvers' card choice (equation 1), revolving (equation 2), and default (equation 3). Together with stochastic assumptions on unobservables, the three equations imply a log-likelihood function for observed decisions.

The log-likelihood $\log \mathcal{L}_{mt}$ for channel m and month t is the sum of the log-likelihood for card choice $\log \mathcal{L}_{mt,E}$ (equation D.1), and the joint log-likelihood for borrowing and default choices $\log \mathcal{L}_{mt,BD}$ (equation D.3). The former is standard for mixed logit models. The latter is complicated by the truncation in borrowing. Specifically, I observe the constrained level of borrowing $b_{ijmt} = \min\{b_{ijmt}^*, \bar{b}_{ijmt}\}$, rather than the desired borrowing, b_{ijmt}^* . Hence, I do not observe the desired borrowing for full-utilization revolvers, which implies that their contribution to the likelihood is an integral with no closed form. I overcome this challenge by using simulated maximum likelihood estimation.

To make likelihood estimation feasible, I specify the initial borrowing level as a function of the advertised interest rate rather than the individualized rate. There are economic and econometric grounds for this simplification, which I detail in Appendix C.1. Most pertinently, individualized rates are revealed some time after approval, so card choice and *initial* borrowing can depend only on the salient advertised rate.

The endogeneity of advertised interest rates in the card choice and borrowing equations must also be addressed for likelihood methods to yield consistent parameter estimates. I sidestep this matter in the likelihood estimation by estimating a full set of product-channel-month fixed effects δ_{jmt} in the card choice and borrowing equations, which subsumes the endogeneity problem created by any correlation between r_{jmt} and ξ_{jmt} .

Formally, I write Equation (1) as $V_{ijmt}^E = \delta_{jmt}^E + \nu_{ijmt} + u_{ijmt}^E$, where

$$\delta_{jmt}^E = \beta^{E'} X_{jmt}^E + \xi_{jmt}^E + \eta_{jmt}^E + \alpha^E r_{jmt}, \quad (9)$$

and $u_{ijmt}^E = \Omega_{mt}^{E,r} \tilde{y}_{imt} r_{jmt}$ (equal to zero if $\Omega_{mt}^{E,r} = 0$). Similarly, I write Equation (2) as $\log(b_{ijmt}^*) = \delta_{jmt}^B + \varepsilon_{imt}^B + u_{ijmt}^B$, where

$$\delta_{jmt}^B = \beta^{B'} X_{jmt}^B + \xi_{jmt}^B + \alpha^B r_{jmt} + \eta_{jmt}^B, \quad (10)$$

and $u_{ijmt}^B = \Omega_{mt}^{B,\text{cons}} \tilde{y}_{imt} + \Omega_{mt}^{B,r} \tilde{y}_{imt} r_{jmt}$. The terms δ_{jmt}^E and δ_{jmt}^B are the card-channel-month fixed effects that are estimated in the first step market-by-market simulated maximum likelihood estimation, along with the variance-covariance matrix of private characteristics $(\varepsilon_{imt}^B, \varepsilon_{imt}^D)$, specifically σ_{mt}^B and σ_{mt}^D , and the demographic coefficients $(\Omega_{mt}^{E,r}, \Omega_{mt}^{B,r}, \text{ and } \Omega_{mt}^{B,\text{cons}})$. Finally, because of the typical identification issue in discrete choice models, I normalize the first value of δ^E in each market to zero, and take interest rates and card characteristics in (9) as differences from the outside option.

Log-likelihood for Borrowing and Transacting. In the second step of demand estimation (Step 2 in Figure A.4), I maximize a log-likelihood for the choice between transacting and revolving, which estimates δ_{0mt} and the transacting utility term $\Omega_{mt}^{E,\text{cons}}$, along with the nesting parameter for the extreme value shocks, ϱ_{mt} .

Constant Demand Parameters. In the third step, I recover the constant parameters of the card-choice and borrowing equations by projecting the estimated fixed effects $(\hat{\delta}_{jmt}^E, \hat{\delta}_{jmt}^B)$ onto market fixed effects, interest rates, and observed characteristics, as in (9) and (10). At this point, I must overcome the correlation between interest rates r_{jmt} and unobservable characteristics ξ_{jmt} , which I do using an instrumental variables approach. The instrument exploits a 2011 cost shock arising from the mis-selling of payment protection insurance (PPI), a product UK lenders bundled with credit cards to cover repayments in the event of adverse

events such as unemployment or disability. Many policies were mis-sold, and in 2011 a High Court ruling upheld the retrospective application of mis-selling standards, forcing lenders to reopen claims. Around 64 million policies were mis-sold between the 1970s and 2000s, and complaints ultimately produced over £38bn (\$60bn in 2015 USD) in repayments (see Appendix D.3 for the institutional background and timeline of events). Because these liabilities fell on each lender in proportion to how heavily it had sold PPI, they raised marginal costs unevenly across lenders. In mid-2011, some lenders responded by increasing advertised rates on their cards. I construct the instrument by interacting lender fixed effects with a post-ruling dummy, so that identification comes from lender-specific changes in advertised rates after the ruling. The identifying assumption is that the ruling shifts card choice and borrowing only through advertised rates. I include lender fixed effects to absorb time-invariant differences across lenders, and I know of no other contemporaneous shock to lenders' unobserved card characteristics in the period I study.

5.2 Supply Estimation

The fourth step of estimation relates to the screening technologies. The parameters to estimate are the screening technology signals $e_{\ell t}$ and the standard deviation of the signal noise, $\sigma_{\ell t}$. I estimate these by minimizing the residual sum of squares from the first order condition of the credit limit optimization problem. Conditional on observables, borrowers who receive the same credit limit from a given lender must have received the same signal; hence, the first-order conditions provide repeated restrictions that let me separately identify the signals and the lender's screening dispersion.

The fifth step of estimation involves estimating the parameters governing the costs of individualizing interest rates, $C(z_{ijmt}; \kappa_{ijmt})$. For this, I specify $C = \frac{1}{2}\kappa_{ijmt}z_{ijmt}^2$ so that the costs of individualizing interest rates are quadratic in the deviation from the advertised rate. From Equation (8), for those with $z_{ijmt} > 0$, $\kappa_{ijmt} = \frac{\partial \Pi_{ijmt}}{\partial z_{ijmt}} \frac{1}{z_{ijmt}}$. Hence, I can estimate κ_{ijmt} for those with $z_{ijmt} > 0$. All parameters in $\partial \Pi_{ijmt} / \partial z_{ijmt}$ are estimated in prior steps.

Its closed-form expression is given in Appendix D.5.

6 Model Estimates and Findings

Below, I discuss the model estimates for the baseline case with $\Omega_{mt}^{E,r} = 0$. Findings are qualitatively similar in the more general version of the model in which $\Omega_{mt}^{E,r}$ is estimated (see Table A.3 for first- and second-step estimates of the general model).

I also experimented with a model that sets both $\Omega_{mt}^{E,r} = 0$ and $\alpha^E = 0$ so that interest rates do not affect card choice at all, and an even more restrictive model with random matching at the card choice stage ($\bar{V}^E = 0$). The latter model cannot match the distribution of market shares in the data, and the model-implied distribution of card-level interest rates set by the lenders in these alternatives is above 90%, far from those in the data, which are around 18–20%. In essence, some consumer aversion to interest rates in the shopping stage is necessary to incentivize lenders to set interest rates to match those in the data.

6.1 Estimates of Demand Parameters

Table 1 presents the means (across the 126 markets) of the demand estimates from the first and second stages of demand estimation. I start with the gradients of default utility, transacting utility, and the level of borrowing with respect to income. Higher-income borrowers are less likely to default ($\Omega^D < 0$), desired borrowing is increasing but concave in relative income ($\Omega^{B,\text{cons}} = 0.38$), and higher-income individuals are less likely to transact ($\Omega^{E,\text{cons}} < 0$).

A key parameter is $\Omega^{B,r}$, which governs how the sensitivity of borrowing to the interest rate varies with income (recall $\alpha_{imt}^B = \alpha^B + \Omega_{mt}^{B,r} \tilde{y}_i$). My negative estimate implies that lower-income borrowers are less sensitive to interest rates (i.e., more inelastic) when deciding how much to borrow, consistent with Nelson (2025) and similar to the finding in Hastings et al. (2017). This estimate creates an incentive for lenders to set higher interest rates for lower-income individuals. Because low-income individuals are more likely to default, the negative

TABLE 1. FIRST AND SECOND STEP DEMAND ESTIMATES

Variable	Interpretation	Parameter	SE ($\times 10^{-2}$)
η^D	Default Equation Constant	-1.88	1.50
Ω^D	Default-Income Gradient	-0.13	1.48
σ^D	Variation in Default Unobservables	0.26	1.52
$\Omega^{B,\text{cons}}$	Revolving-Income Gradient	0.38	1.47
$\Omega^{B,r}$	Income Gradient for Revolving Sensitivity	-0.89	1.43
σ^B	S.D. in Revolving Unobservable	1.81	3.15
$\text{Corr}(\varepsilon^B, \varepsilon^D)$	Correlation in Unobservables	0.19	1.52
$\Omega^{E,\text{cons}}$	Transacting-Income Gradient	-0.10	1.38
ϱ	Transact/Revolve Substitution Parameter	0.28	0.30

Notes: This table reports mean parameter estimates and their standard errors across markets. These estimates are from the baseline model with $\Omega_{mt}^{E,r}$ set to zero. See Table A.3 for first- and second-step estimates of the general model.

correlation between credit scores (e.g., FICO) and interest rates in other settings may reflect price discrimination as well as the pricing of default risk.

Regarding the distribution of private characteristics, the mean correlation between unobserved preferences for borrowing and for default is 0.19, implying adverse selection along the intensive borrowing margin: revolvers with a higher unobserved preference for higher borrowing also have a higher unobserved preference to default. The estimate is slightly larger than the estimate of 0.14 for Italian small business loans (Crawford et al., 2018).

Next, I assess model fit. Figure A.5 shows the model-implied and data distributions of market shares and borrowing. The fit for market shares is very tight, indicating that the model captures heterogeneity in card-choice behavior. In the baseline with $\Omega_{mt}^{E,r} = 0$, the model fit is acceptable but cannot produce the full variation in borrowing in the data. In the version of the model with $\Omega_{mt}^{E,r}$ estimated, the fits of the borrowing and market share distributions are both tight.

Table A.4 reports estimates for the demand parameters recovered in the third stage of demand estimation. OLS coefficients on interest rates in both the card choice (α^E) and borrowing (α^B) equations are positive, whereas IV estimates are negative, indicating interest rate

endogeneity. Coefficients on dummies for air miles and affinity rewards in the card choice equation are positive, with a stronger preference for air miles than affinity rewards.

Lastly, I turn to interest rate elasticities. In the baseline model, α_i^E does not vary with income. Instead, the income-varying response operates through borrowing, so I focus on borrowing elasticities. Figure A.6 plots the density of borrowing elasticities (Equation (C.3) in Appendix C.7) over individuals. On average, demand for credit card borrowing is inelastic, with a mean of -0.20, and there is large dispersion across individuals: the coefficient of variation of the elasticity is 93%. The distribution is skewed, with a large mass near zero, implying that several consumers are almost completely inelastic to the interest rate. Elasticities are relatively similar to other estimates of interest rate elasticities in alternative credit markets (Karlan and Zinman, 2018) and in the US business credit card market (Benetton and Buchak, 2024).

6.2 Estimates of Supply Parameters

Two main findings emerge from estimating the screening technologies (for interested readers, I present additional results in Appendix E). First, signal mismeasurement varies across lenders, σ_ℓ , and screening technology signals, e_ℓ . The coefficient of variation in σ_ℓ across lenders is 68%. Second, the correlation between σ_ℓ and the proportion of periods in which individuals repay the entire balance is 0.54. This estimate is consistent with a segmentation of credit card lenders in which those with the most precise screening technologies serve a riskier but, on average, more profitable market segment.

I also obtain estimates of the costs of individualizing interest rates. Across those with a deviation from the advertised rate, the mean value of κ is 16,430, implying that increasing a single individual's rate ten percentage points above the advertised rate implies an average cost of £123 (\$195 in 2015 USD). Figure A.7 shows the density of κ_{ijmt} estimates, revealing substantial variation, with values as high as 60,000 for some individuals. Given the hetero-

generity in $\Omega^{B,r}$, a wide range of κ values allows the model to rationalize that lenders do not fully exploit the regulation’s flexibility to individualize interest rates.

7 Counterfactual Analysis

Now, I take the estimated model and use it to investigate how the credit card market would be impacted when the two key instruments—interest rates and credit limits—are either allowed to be fully personalized or held constant across customers at the product level. I re-solve the lender profit maximization problem when lenders may individualize interest rates, or credit limits, or both, or neither, and, in doing so, answer three related questions. First, which margin of flexibility is more valuable to lenders: individualized interest rates or individualized credit limits? Second, what are the effects of enabling fully personalized interest rates alongside already personalized credit limits? Third, how much can lenders substitute for tailored interest rates by individualizing credit limits instead?

Implementation. To implement each of these counterfactual scenarios, I take the November 2011 in-branch market (the one for which the parameter estimates are closest to the average estimates over markets), re-solve the lenders’ optimization problem under the new regime, and then measure three sets of endogenous variables.⁸ The first set comprises the equilibrium origination interest rates and credit limits. The second set comprises the consumer’s card choice, a dummy for whether they revolve or transact, and their borrowing level. The third set comprises the supply-side variables. I focus on profit, with the lender’s profit from revolver i given by

$$\pi_{ij}^{\text{post}} = b_{ij} \left[(1 - \mathcal{D}_i)(r - c_j) + \mathcal{D}_i(-1 - c_j) \right],$$

where $\mathcal{D}_i = 1$ if borrower i defaults and 0 otherwise.

⁸Results obtained from alternative choices for the counterfactual market are similar.

7.1 Interest Rate Flexibility versus Credit Limit Flexibility

In the first exercise, I quantify the relative merits to lenders of standalone interest rate and credit limit flexibility. I compare the baseline case, in which interest rates are constant at the card level and credit limits are personalized, to a counterfactual in which interest rates are personalized and credit limits are fixed at the card level. To construct the counterfactual regime, I have lenders set card-level credit limits and set the cost of individualizing interest rates κ_{ijmt} to zero and remove the regulatory constraint in Equation (6), thereby allowing unconstrained interest rate individualization.⁹ Appendix F formalizes and describes my implementation of the lender’s problem in this counterfactual and the other two to follow.

The primary finding is that interest rate flexibility is the more valuable of the two margins to lenders: a lender free to individualize interest rates rather than credit limits earns approximately 4% more on average than in the credit limit-only baseline. Hence, lenders value the margin not targeted by regulation only slightly less than the one that is targeted.

That interest rates are preferred is a feature of the model estimates, not of the model’s structure. At the estimated demand elasticities (in which customers have inelastic demand on average), when lenders can costlessly personalize interest rates, they can raise the rate charged to a risky borrower without a high chance of losing them to transacting or to a rival. But if shopping-stage demand were sufficiently elastic, for example, this mechanism would fail. Such a borrower, faced with a higher rate, would either transact or switch lenders, and credit limits would become the more valuable instrument, since they can still be used to

⁹Because I remove the advertised-rate requirement, a consumer choosing a card in this regime observes their own lender-specific interest rate rather than an advertised one. Also, I carry over from the estimated demand model the assumption that they do not observe their credit limit at the choice stage. This informational assumption applies to every counterfactual regime. I maintain the assumption on credit limits primarily because potential preferences over credit limits in counterfactuals cannot be recovered. Also, the lender problem in this case is infeasible to solve exactly; Appendix F.3 describes the approximation.

control a risky customer’s borrowing.¹⁰

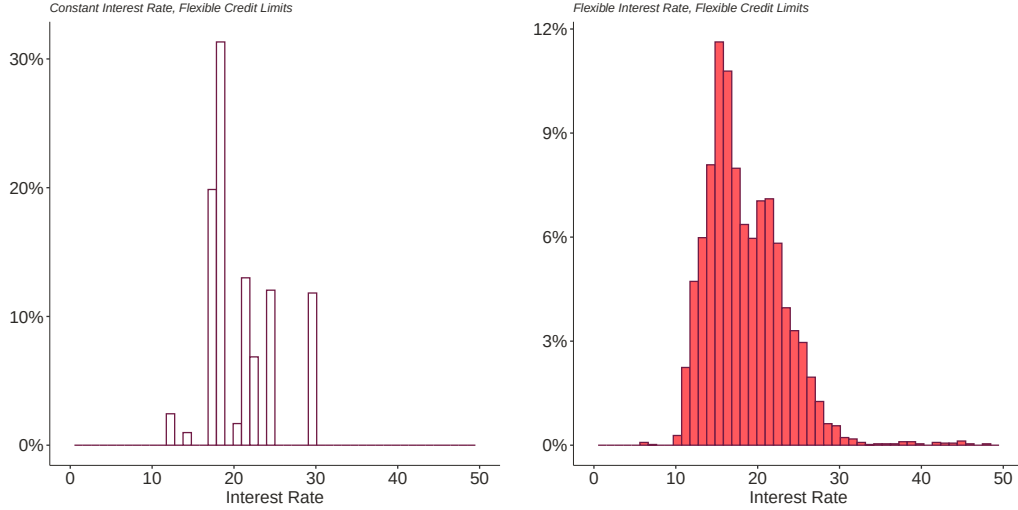
7.2 Personalizing Interest Rates and Credit Limits

My second counterfactual exercise allows lenders to individualize both interest rates *and* credit limits without costs or regulatory constraints. I compare this scenario to the baseline in which lenders individualize credit limits only. This exercise lets me explore whether adding interest rate flexibility to credit limit flexibility raises lender profits. A monopolist cannot be worse off with extra flexibility, as the card-level outcome is always still available. However, whether adding price personalization raises or lowers lender profit in an oligopoly is not obvious *a priori*: Rhodes and Zhou (2024) shows that the profit effect of enabling personalized pricing under imperfect competition can go either way, depending on the degree of market coverage, which can be thought in my context of as the proportion of potential customers already served in the revolving nest (that is, customers not transacting). The key intuition is that when lenders compete, greater pricing flexibility lets them attract consumers who would otherwise not revolve a balance, but it also intensifies competition for each individual customer, including those already served. The headline result is that the benefit from price discrimination dominates: full personalization raises lender profit by 12% relative to the baseline with tailored credit limits but constant rates.

Figure 3 shows the distribution of interest rates in the baseline and the counterfactual scenario. The distribution becomes highly individualized in the counterfactual, with many unique interest rate values compared to a few distinct values in the baseline. The coefficient of variation in interest rates increases by 27% and the standard deviation increases by 13%. Together, these statistics imply a marked increase in interest rate dispersion when costs and constraints are removed.

¹⁰In a comparative-statics exercise, I hold all other parameters fixed and set the shopping-stage interest rate sensitivity parameter α^E around 25% lower, to -2 . In this case, credit limit flexibility is more valuable, with profits approximately 13% higher under credit limit flexibility than under interest rate flexibility.

FIGURE 3. DISTRIBUTION OF INTEREST RATES IN DATA AND COUNTERFACTUAL



As anticipated, interest rates rise for low-income customers and fall for high-income customers. Credit limits increase once interest rates can be fully personalized, with the share of borrowers who exhaust their limit falling close to zero. Borrowing rises for high-income customers because of lower interest rates. Average borrowing also rises for low-income customers (who were often credit constrained in the baseline) with the relaxation of previously binding credit limits, despite the higher interest rates. Lenders can allow low-income, risky customers to borrow more and still be profitable now that they are able to price in the customer’s risk through the interest rate.

Why does full personalization raise average profit relative to the baseline in this case? In line with Rhodes and Zhou (2024), *revolving* is not a full-coverage segment—a large number of cardholders transact rather than revolve (recall from Section 2 that around half of all customers pay their balance in full each month). In the counterfactual, fewer customers transact relative to the baseline, especially higher-income transactors who are incentivized into revolving through lower interest rates. Furthermore, the heterogeneity in borrowing demand elasticities, coming from $\Omega^{B,r}$ being estimated to be strongly negative, offers a mechanism through which lenders can influence the borrowing behavior of customers of varying demand elasticity through tailoring interest rates: were $\Omega^{B,r}$ close to zero, this would

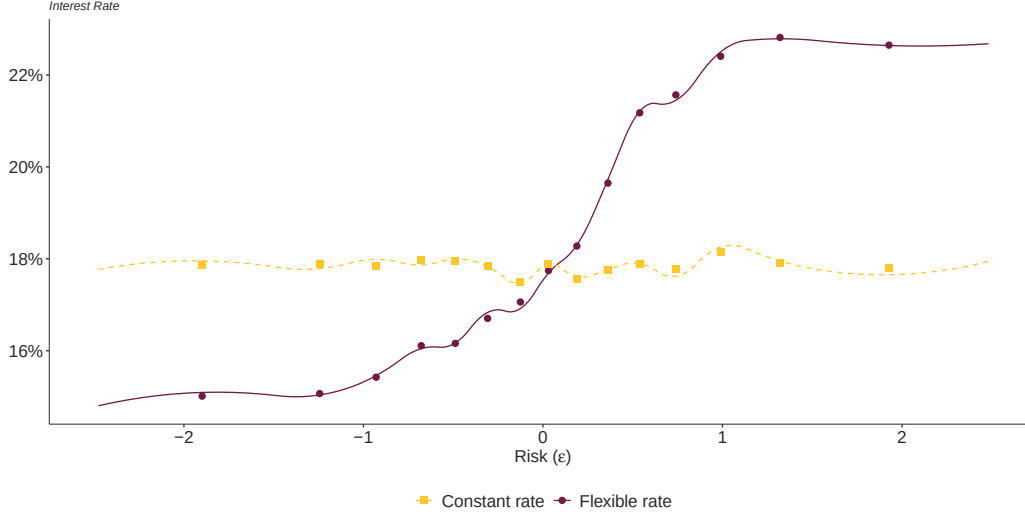
not be possible.

Although personalizing interest rates alongside personalized credit limits increases average profits, some lenders earn less in the counterfactual, and the value of the two instruments combined is less than the sum of their separate values. To show this latter result, I simulate a third counterfactual scenario in which lenders are forced to set card-level constant credit limits and interest rates, which I call a “benchmark” counterfactual. Profits are lower in this benchmark than in all other cases. Adding interest rate personalization when credit limits are constant raises profit by 16% relative to the benchmark, but adding it to already-personalized credit limits raises profit by only 12%.

To see this another way, let $\Pi_{x,y}$ denote total profits for interest rate regime $x \in \{pr, cr\}$ (personalized rates or constant rates) and credit limit regime $y \in \{pcl, ccl\}$. I then calculate $\mathcal{I} = (\Pi_{pr,pcl} - \Pi_{cr,pcl}) - (\Pi_{pr,ccl} - \Pi_{cr,ccl})$. This quantity represents the benefits of adding personalized pricing to personalized credit limits, *relative* to the benefits of adding personalized pricing to constant credit limits. Because $\mathcal{I} < 0$, in this sense, the two tools are submodular rather than supermodular: each tool is worth less at the margin once the other is in use. The formula can be rearranged to also show that the benefit of flexible credit limits is lower when interest rates are flexible than when they are fixed.

Finally, I illustrate the full extent of risk-based pricing in the flexible interest rate and credit limit counterfactual. Figure 4 shows a binscatter plot of origination rates against the risk variable ε in the fully flexible counterfactual and the benchmark constant rate counterfactual (the plot is similar when the baseline case is used rather than the benchmark). Interest rates in the benchmark case (the dashed golden line with square points) are flat across the range of values for ε . In contrast, interest rates increase with risk in the counterfactual case (solid maroon line with circular points). The lines cross at 0, the mean of the distribution of ε , implying higher rates for individuals with above-mean risk in the fully flexible counterfactual and lower rates for below-mean risk individuals.

FIGURE 4. COUNTERFACTUAL INTEREST RATES BY RISK



Notes: “Constant rate” refers to the benchmark counterfactual in which lenders must set card-level interest rates and credit limits. “Flexible rate” refers to the counterfactual in which lenders are unconstrained and can set personalized interest rates and credit limits.

7.3 Offsetting Rate Regulation with Credit Limits

Finally, I quantify how much of the restrictions on personalizing interest rates can be offset through personalizing credit limits. This question is of practical importance because most credit card regulation constrains the interest rate, and lenders in the UK choose to personalize contracts through credit limits.

I find that when interest rate tailoring is constrained, lenders recover a substantial share of the profit that full personalization would deliver through adjusting on the credit limit margin. Relative to the benchmark in which neither instrument is individualized, tailoring credit limits alone raises profit by 11%, compared with the 24% that individualizing both instruments delivers (see Table 2 for the full set of percentage changes). Credit limit flexibility on its own thus recovers roughly 46% of the total gain from personalizing interest rates and credit limits. Credit limits are therefore an important instrument in their own right: a lender facing costs and constraints on tailoring interest rates can claw back close to half of the profit

TABLE 2. LENDER PROFIT UNDER PERSONALIZATION OF EACH INSTRUMENT

	Card-level Interest Rate	Personal Interest Rate
Card-level Credit Limit	0%	+16%
Personal Credit Limit	+11%	+24%

Notes: Each cell reports the percentage change in aggregate lender profit relative to the benchmark in which lenders individualize neither instrument (top-left).

it would otherwise forgo through adjusting credit limits.

Assumptions Underlying the Counterfactual Exercises. As is standard, these counterfactual scenarios assume that all exogenous model elements remain fixed. In practice, product menus may shift when restrictions on individualizing interest rates and credit limits are introduced or removed, and lenders may also adapt their screening technologies to the new environment. Beyond the credit card market, my framework cannot account for endogenous responses in adjacent markets—personal loans or Buy Now Pay Later, say—that might respond to changes in the credit card market and feed back into consumer and lender behavior. Finally, the intuitions behind these results need not carry over to other credit markets in which product choice itself depends on the quantity of credit a borrower requires. A complete assessment of the net effects of altering interest rate and credit limit flexibility would need to accommodate these margins.

8 Conclusion

In this paper, I investigate how credit card lenders individualize contracts through credit limits and interest rates. I use novel, statement-level microdata to show that UK lenders personalize credit limits but set near-uniform interest rates across customers with varying default risk. I then estimate a structural model of the credit card market in which lenders endogenously set risk-based credit limits alongside interest rates, subject to UK regulatory requirements that constrain interest rate personalization. The model features credit rationing

on the intensive margin as lenders set binding credit limits based on predictions of customers' unobserved risk.

I use the estimated model to answer three questions through counterfactual analyses. First, when offered the choice between personalized interest rates or credit limits, lenders slightly favor the former, earning 4% higher profits than under credit limit flexibility. Second, adding interest rate flexibility on top of credit limit flexibility raises profits by 12%, a result consistent with the predictions of prior theoretical work on the profit effects of price personalization, given the moderate coverage of the credit card revolving market and heterogeneous borrowing demand elasticities. Third, when personalized pricing is constrained, the ability to personalize credit limits allows lenders to recoup 46% of the profits that would otherwise be lost.

Questions remain about the sources of the costs of individualizing prices and why UK credit card lenders do not offer a menu of cards with varying interest rates or reprice customers after origination. Appendix C.5 considers some candidate explanations, but answering these questions is a fruitful avenue for further work. Regarding external validity, consumer credit markets across countries and over time have used varying combinations of individualized prices and quantities. No general theory explains how product features and regulatory environments interact to influence how lenders individualize contracts. Deriving market-level conditions that yield tailored prices, quantities, or both is a challenge for future research.

References

- AGARWAL, S., S. CHOMSISENGPHET, N. MAHONEY, AND J. STROEBEL (2014): “Regulating Consumer Financial Products: Evidence from Credit Cards,” *The Quarterly Journal of Economics*, 130, 111–164. 5
- (2017): “Do Banks Pass through Credit Expansions to Consumers Who want to Borrow?” *The Quarterly Journal of Economics*, 133, 129–190. 4, 12

- AYDIN, D. (2022): “Consumption Response to Credit Expansions: Evidence from Experimental Assignment of 45,307 Credit Lines,” *American Economic Review*, 112, 1–40. 4, 17
- BENETTON, M. AND G. BUCHAK (2024): “Revolving Credit to SMEs: The Role of Business Credit Cards,” *Working Paper*. 7, 22, 28
- BERGEMANN, D., B. BROOKS, AND S. MORRIS (2015): “The Limits of Price Discrimination,” *American Economic Review*, 105, 921–57. 5
- BERGER, A. N., C. H. S. BOUWMAN, L. NORDEN, R. A. ROMAN, G. F. UDELL, AND T. WANG (2024): “Piercing through Opacity: Relationships and Credit Card Lending to Consumers and Small Businesses during Normal Times and the COVID-19 Crisis,” *Journal of Political Economy*, 132, 484–551. 7
- BORD, V. M., A. KOVACS, AND P. MORAN (2025): “Automated Credit Limit Increases and Consumer Welfare,” Finance and Economics Discussion Series 2025-088, Board of Governors of the Federal Reserve System. 7
- CALOMIRIS, C. W., S. D. LONGHOFFER, AND D. M. JAFFEE (2017): *Credit Rationing*, London: Palgrave Macmillan UK, 1–10. 4
- CORTS, K. S. (1998): “Third-Degree Price Discrimination in Oligopoly: All-Out Competition and Strategic Commitment,” *The RAND Journal of Economics*, 29, 306–323. 5
- CRAWFORD, G. S., N. PAVANINI, AND F. SCHIVARDI (2018): “Asymmetric Information and Imperfect Competition in Lending Markets,” *American Economic Review*, 108, 1659–1701. 13, 22, 27
- DRECHSLER, I., H. JUNG, W. PENG, D. SUPERA, AND G. ZHOU (2025): “Credit Card Banking,” Staff Report 1143, Federal Reserve Bank of New York. 7
- DUBE, J.-P. AND S. MISRA (2023): “Personalized Pricing and Consumer Welfare,” *Journal of Political Economy*, 131, 131–189. 5

- DURRMEYER, I., J.-F. FOURNEL, AND M. SAMANO (2026): “Personalized Pricing and the Value of Past Purchase Histories: An Empirical Perspective,” *Working Paper*. 5
- EVANS, D. AND R. SCHMALENSSEE (2005): *Paying with Plastic: The Digital Revolution in Buying and Borrowing*, MIT Press Books, The MIT Press, 2nd ed. 13, 19, 20
- FCA (2015a): “Credit Card Market Study Interim Report: Annex 10 - Account Level Data,” *Financial Conduct Authority Research Publications*. 6
- (2015b): “Credit Card Market Study Interim Report: Annex 3 - Results from the consumer survey,” *Financial Conduct Authority Research Publications*. 15
- (2024): “CONC 5.2A: Creditworthiness Assessment,” FCA Handbook, Consumer Credit Sourcebook (CONC). 16
- GALENIANOS, M. AND A. GAVAZZA (2022): “Regulatory Interventions in Consumer Financial Markets: The Case of Credit Cards,” *Journal of the European Economic Association*, 20, 1897–1932. 4, 9
- GATHERGOOD, J., N. MAHONEY, N. STEWART, AND J. WEBER (2019): “How Do Individuals Repay Their Debt? The Balance-Matching Heuristic,” *American Economic Review*, 109, 844–75. 14
- GROSS, D. B. AND N. S. SOULELES (2002a): “Do Liquidity Constraints and Interest Rates Matter for Consumer Behavior? Evidence from Credit Card Data,” *The Quarterly Journal of Economics*, 117, 149–185. 4
- (2002b): “An Empirical Analysis of Personal Bankruptcy and Delinquency,” *The Review of Financial Studies*, 15, 319–347. 17
- HASTINGS, J., A. HORTAÇSU, AND C. SYVERSON (2017): “Sales Force and Competition in Financial Product Markets: The Case of Mexico’s Social Security Privatization,” *Econometrica*, 85, 1723–1761. 26

HERKENHOFF, K. AND J. M. MORELLI (2026): “Credit Card Market Power,” Working paper, SSRN. 7

HOLMES, T. J. (1989): “The Effects of Third-Degree Price Discrimination in Oligopoly,” *The American Economic Review*, 79, 244–250. 5

KARLAN, D. AND J. ZINMAN (2018): “Long-Run Price Elasticities of Demand for Credit: Evidence from a Countrywide Field Experiment in Mexico,” *The Review of Economic Studies*, 86, 1704–1746. 28

LANCASTER, K. J. (1966): “A New Approach to Consumer Theory,” *Journal of Political Economy*, 74, 132–157. 13

LIVSHITS, I., J. C. MAC GEE, AND M. TERTILT (2016): “The Democratization of Credit and the Rise in Consumer Bankruptcies,” *The Review of Economic Studies*, 83, 1673–1710. 4

NELSON, S. T. (2025): “Private Information and Price Regulation in the US Credit Card Market,” *Econometrica*, 93, 1371–1410. 1, 4, 5, 10, 17, 26

POLO, A., A. TABURET, AND Q.-A. VO (2025): “Screening using a menu of contracts: A structural model for lending markets,” *Journal of Financial Economics*, 169, 104056. 4

RHODES, A. AND J. ZHOU (2024): “Personalized Pricing and Competition,” *American Economic Review*, 114, 2141—70. 5, 31, 32

SHAFFER, S. (1998): “The Winner’s Curse in Banking,” *Journal of Financial Intermediation*, 7, 359–392. 19

STATE OF CONNECTICUT (2026): “An Act Concerning Consumer Privacy and Protection,” Substitute S.B. 4 (surveillance-pricing disclosure provisions), 2026 Conn. Pub. Act No. 26-64. 3

STATE OF MARYLAND (2026): “Protection From Predatory Pricing Act (H.B. 895),” 2026 Md. Laws ch. 154. 3

STATE OF NEW JERSEY (2026): “Fair Price Protection Act (A. 4085/4523),” P.L. 2026, c. 65. 3

STIGLITZ, J. E. AND A. WEISS (1981): “Credit Rationing in Markets with Imperfect Information,” *American Economic Review*, 71, 393–410. 4

THISSE, J.-F. AND X. VIVES (1988): “On The Strategic Choice of Spatial Price Policy,” *The American Economic Review*, 78, 122–137. 5

YIN, X. (2026): “Learning in the Limit: Income Inference from Credit Extensions,” *The Journal of Finance*, Forthcoming. 4